Of Earth and Sun: Generative Soundscape Composition and Biophilic Design

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ABSTRACT

Despite well-documented benefits experienced by communities and individuals with easy access and direct exposure to nature, many individuals spend a majority of their working hours indoors. The field of sustainable design has tackled this issue through biophilic design, which strives to elicit a positive, valued experience of nature in the human built environment. But while biophilic design principles are increasingly employed within the visual domain, auditory applications of these principles are underutilized and underexplored. I examine sonic approaches to biophilic design in my generative soundscape installation, *Of Earth and Sun.*

In 2013, the Phipps Conservatory and Botanical Gardens in Pittsburgh, PA, commissioned *Of Earth and Sun*, a permanent sound installation for the public atrium of the Center for Sustainable Landscapes (CSL).

This project is part of its Biophilia Enhanced Through Art (BETA) initiative, which uses art to remind people about nature's beauty and the connections between humans and the natural world. *Of Earth and Sun* is a dynamic sound installation that evolves throughout the day and with the seasons. The systems at the CSL respond to environmental input in order to reduce its ecological footprint. Similarly, *Of Earth and Sun* uses data from the CSL's on-site weather station to dynamically control the sounds and processes that will create the installation. Sounds and soundscapes gathered from throughout the Pittsburgh region are stored in a local database, processed, and played back through transducers placed on windows throughout the CSL's atrium.

I am installing the fourth and final iteration of this project in the late summer and early fall of 2016. In this paper, I describe the project goals, processes, and outcomes of the various project iterations. I also examine listener experience and propose models for enhancing listener engagement in soundscape composition using diverse models of community engagement and user interface design.

Biosphere Soundscapes: Exploring the art and science of listening to UNESCO Biosphere Reserves

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ABSTRACT

Biosphere Soundscapes is a large-scale interdisciplinary research project underpinned by the creative possibilities of acoustic ecology, ecoacoustics and rapidly emerging fields of biology concerned with the study of environmental patterns and changes through sound. This project is designed to inspire communities across the world to listen to the environment and explore the value of sound as a measure for environmental health in UNESCO biosphere reserves. The project is delivered through immersive residencies with artists and scientists, research laboratories, intensive masterclasses and a diversity of creative projects spanning four continents.

Biosphere reserves are sites recognised under UNESCO's Man and the Biosphere Program (MAB) to promote innovative approaches to sustainable development. There are currently 669 biosphere reserves in 120 countries comprising terrestrial, marine and coastal ecosystems. Each biosphere reserve is designed and managed in a different way, but all seek to reconcile the conservation of biological and cultural diversity.

They differ from world heritage sites in that they encourage active community participation and are ideal locations to test and demonstrate innovative approaches to ecosystem monitoring and sustainable development.

Biosphere Soundscapes draws on the inherently interdisciplinary nature of sound to explore cultural and biological diversity through accessible audio recording technologies, interdisciplinary creativity and environmental engagement with local and global communities.

This paper introduces the framework and methodology for Biosphere Soundscapes and explore the ecological, social and cultural contexts of UNESCO Biosphere Reserves through sound. This presentation will also introduce the potential role of acoustic ecology in the Lima Action Plan (2016-2025) adopted by UNESCO at the 4th World Congress of Biosphere Reserves in Lima, Peru in March 2016. Biosphere Soundscapes sits at the intersection of art and science, with the recordings providing valuable scientific data for biodiversity analysis and incredible source material for creative works that can bring awareness to these environments through new technologies. This project is designed as a platform for artists, scientists and global communities to collaborate and expose the creative and scientific possibilities of environmental sound to a global audience.

Virtual Reality, Spatial Sound and the Future of Ethnomusicology: The *Music Legacy Project*

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ABSTRACT

Since SXSW (*South by Southwest* in Austin, Texas) 2014's edition and the presentation of "Strangers with Patrick Watson", produced by Felix & Paul Studios, cinematic films in 3D and 360° are about to become a major reality for the music industry. With the commercialization of virtual reality head-mounted displays (Oculus Rift, Samsung Gear VR, HTC Vive...), demand for creative content for this new way of listening to music performances opens Pandora's box for every music maker – and also for any ethnographer.

By putting the user at the center of an immersive experience, VR technology offers new ways to think about the relationship between the artists and their audiences. From the research side, how this paradigmatic shift could influence the way we do music ethnography? What happens with the researcher's position? What about the musicians? How to measure the artists' choices in the data analysis? How to manage the marketing of such content?

After an overview of some audio recordings techniques used in virtual reality, this communication will present our first ethnomusicological data collected with a 360° audio-visual recording device. This will allow us to estimate the role of sound recording in the "restitution" in VR. From a presentation of our postdoctoral research, the *Music Legacy Project* (www.musiclegacyproject.com, developed in partnership with the audio-visual post-production company La Hacienda Creative, based in Montreal and Toronto, Canada) – this paper will also address some of the methodological, ethical and epistemological issues raised by the use of such a experimental device.

LISTENING TO DESERTS IN THE AMERICAN SOUTHWEST: GARTH PAINE'S EXPLORATIONS OF SONIC PLACEMAKING

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ABSTRACT

Thanks to their fierce nature and potential for metaphor, deserts have long fascinated musicians. Edgard Varèse, Olivier Messiaen, Luc Ferrari and Peter Sculthorpe created works inspired by deserts around the world, but they had only a tenuous connection with these places. In contrast, David Dunn, Richard Lerman, Maggi Payne and Garth Paine have strongly identified with deserts and paid tribute to them in numerous works, compelling examples of sensitive engagement with these places. This paper centers on Australian-born composer and sound artist Paine. I will analyze and contextualize his large-scale interdisciplinary and collaborative Listen(n) Project, his acousmatic work *Becoming Desert* (2014) and live-electronic flute piece *Forest* (2015) which have been inspired by deserts in the American Southwest and draw on field recordings made with ambisonic recording technology. I will explain how they reflect Paine's environmental philosophies and concepts of sonic placemaking in the context of composition, virtual reality experiences, community art and citizen science projects.

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RE-IMAGINING A SITE SPECIFIC SOUNDSCAPE DESIGN PROJECT INTO A SONIC ART PERFORMANCE SPACE

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ABSTRACT

Over the past 20 years or more I have been mostly involved the world of design, creating functional soundscapes in public spaces; zoos, memorials, architectural precincts and more. To assist my own comprehension of my activities I contend that *design* is about working with materials and *art* is about working with ideas. This distinction, while artificial and notional, recognises the very different qualities and performance parameters pertaining to each domain. This presentation intends to exemplify these differences in a discussion about the creation of the sound work, 'What U might have heard..'

1. INTRODUCTION

'What U might have heard..', the Sound Work, is a re-envisioned version of an ambient electroacoustic soundscape installation for the public areas of the Australian Centre for the Moving Image (ACMI Soundscape), Melbourne, in 2000. While not what I would consider a pure work of *art* the piece does go some way in that direction once emancipated from its functional role at ACMI.

The original project was a site specific work carefully designed into this unique precinct and public space. It functions in a number of ways; to help orientate visitors within a highly dynamic architectural space, as an aesthetic overlay and to extend the presence or image of ACMI the institution into the acoustic domain. The Sound Work intends to relive the experience of visiting that space through creative composition rather than simply as documentation.

2. CONTENT AND LICENCE TO PERFORM

Any sound heard is a function of time; the fusion of *content* and *performance*. Content can be developed independently and silently however when it is performed in public, in order to be environmentally responsible, all sounds must conform to what might be considered a *licence to perform.*

While the ACMI soundscape may be ever present and able to be heard within the precinct it, by necessity, is not designed to be listened to. Its inherent texture of *background* and *midground* layers can be punctuated occasionally with dynamic *foreground* sounds. Cogniscant of how people are inhabiting the building the interonset times of such 'alerting' events must be programmed carefully. Foreground sounds can be created either as loud(er) sounds or of a character which jumps out of the environment and causes attention. This introduces a somewhat playful mode or interplay between what is real and what is not real. A kind of augmented aural reality.

The same texture is present in 'What U might have heard..' however the licence to perform is completely emancipated from any functional role and responsibilities. The sound materials are free to be heard in much greater detail and across a much wider dynamic range. Without the physical presence of the building the playful sense of what might or might not be real is lost. As will be described later, there is another similar kind of interplay within the sound materials that can be playfully explored.

3. DIFFERENTIATING

In the two versions of the soundscape, let's call them the *design* (ACMI soundscape) and *art* versions (Sound Work), the *content* remains fairly consistent. It is in the *performance* where the two versions are completely differentiated. In the *design* version people are free to move about from place to place thereby creating their own 'performance'. The temporal structure is therefore free and arbitrary. The *art* version would have the audience remaining in one position, perhaps seated in a concert situation, with the piece 'spatialised' around them. In a sense the physical building itself becomes a virtual artefact - or ghost. The temporal structure in this case is predetermined or composed. The *composer* is required to make critical decisions about how time passes. For a sound designer, unused to dictating terms on how a soundscape will be heard, this poses the first of a number of challenges.

4. UNDERLYING SONIC STRUCTURE

The sonic layout plan for ACMI was derived from the orientation of the building on Melbourne's CBD grid, or more generally, on the cardinal directions NESW. Each

cardinal point is characterised by the placement of one of five iconic sound sources or themes. In the North zone where the building fronts the city streetscape, soundmarks of Melbourne such as trams and trains are performed, reflecting the origins of the site as a public transport hub. Sounds derived of nature, reflecting the perspective of the leafy suburbs and nearby hills to the East of the city were sent into the building's eastern atrium. The sounds of water, notionally connecting the wet areas of the building to the nearby river and ultimately the ocean were delivered within the toilets to the West. Sounds reflecting on celebration and festivities perform in the Southern plaza where ACMI meets Federation Square.

The sound of footfall or footsteps representing the habitation of an urban space were delivered into stairwells. And, in recognition of the cultural institution housed within the building, the sound of a hand clap, was the basis for sounds generated for the central atrium - for it has been said that in 1932 an audience erupted into spontaneous applause when they heard the well-known sounds of Australian birds in the soundtrack of one of the first 'talkies' to be shown in Australia, Cinesound's 'On Our Selection'.

This schematic is maintained as a structural element in the Sound Work. Unlike the visitor wandering freely around the building, for the Sound Work the composer must decide and commit to a specific path through this structure. The chosen trajectory begins in the central atrium and stairwell, moves through a transition air lock and out onto the street to the North. The listener is then taken back into the building and down a corridor to the toilets on the ground floor. The trajectory passes back through the atrium and to the three levels of the eastern stairwell and ultimately out to the southern plaza. These decisions are guided by the need to create an over-arching narrative which includes periods of contemplation, transitions, forward impetus and a sense of recapitulation and departure.

5. GENERATING AND PERFORMING THE CONTENT

An image of an acoustic space can be created using the scientific technique of the *impulse response*. A short impulse is sounded in the space and the response is recorded as the sound reflects off the structure and decays over time. For the project 'iconic' sounds of Melbourne, rarely heard in their prime form within ACMI, are departure points (*impulses*) for the creation of the content both in terms of inspiration as well as the actual production of a large library of sound materials. The impulses themselves are very short sounds, a hand clap, a train horn, a tram bell, a bird call or a drip of water each lasting for only a few seconds. An ambient soundscape running for 12 hours a day or more will demand great variety and a non-repetitive *duty cycle*. The *'response'* sounds are the product of extensive DSP manipulation of the impulses using granular synthesis, time stretching, multiple echo and convolution reverb effects.

In the ACMI soundscape a computer programme randomly selects sound files from the library and streams them into the various areas of the precinct. It is highly unlikely that a visitor will hear any sound repeated in the time they will be in the building. It is even less likely that they will hear one of the prime sounds. This unpredictability is essential for maintaining a sense of freshness over months and years of operation. In a 25 minute sound composition such randomness may be desirable for compositional reasons however it is not a requirement.

Within 'What U might have heard..' all of the prime sounds are performed unaltered at some point in the composition. They are embedded into the layers of related processed sounds so that the relationship between the real and unreal can be explored by the listener. In the same way that one may wonder if that tram bell heard out on the street is from an actual tram, one may also question whether those bird like sounds are actually a bird or some other unknown source. The use of granular synthesis is particularly useful for generating a vast array of sounds from a small fragment or prime sound. A single short magpie call can generate sounds across a wide range that are either clearly derivative or almost unrecognisable from the original. This interplay is exploited much more overtly within the 'contemplative' sections of the piece than would ever have been possible in the original soundscape.

In order to create a balanced and well structured composition a number of other devices were adopted. The piece is punctuated with dynamic surprise sounds such as a loud bell, wild applause or sounds with a very wide frequency range. These sounds could never be performed so overtly in a public space such as ACMI. They are, however, perfectly acceptable with the bounds of the concert hall.

The selection and layup of sound files within the digital studio (Digital Performer) was often informed by the rhythmic elements within the sound files. Such control over rhythm would not be possible in the random file selection process in the ACMI soundscape. Any resulting rhythmic structure would be the result of pure coincidence.

Finally, in order to master the piece into an 8 channel format for a concert diffusion system, the output was spatialised using convolution reverb processing. The desire was to attempt to replicate or create a sense of the precinct of ACMI and its soundscape delivered from 48 channels and 120 hidden loudspeakers.

6. CONCLUSION

This paper concentrates on the challenges of moving from the modality of *design* to that of *art*. It has been proposed that many elements such as content may remain essentially unchanged. However the performance of that content will require a whole new set of considerations in order to be successful. The physical realisation of all sound needs to conform to a *performance licence.* Such a licence is derived through careful assessment of the physical and functional requirements of any given space. The freedom of a controlled space such as a concert hall cannot be transferred into a public or urban precinct or that of an institution such as ACMI, Melbourne. A designed soundscape may not perform well as a sound art work without thoughtful reworking of numerous performance parameters.

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What U might have heard.. is a piece commissioned for the RMIT Sonic Arts Collection

RMIT UNIVERSITY

Sonic Environment in Vanuatu: Exploring Water Music TRACES

Sally Ann McIntyre
Independent Artist

ABSTRACT

Huia Transcriptions: re-collecting Colonial era witness accounts of extinct birdsong

In classical acoustic-ecological conceptions of the soundscape, the technological preservation of a sound mark might be understood to positively relate to the preserving of memory of place. But what of the sounds beyond (recorded) memory, that are already missing? Might there be value in suspending the fantasy of a natural plenitude of sonic fecundity, and its status as a potential recorded totality, to adequately acknowledge this haunting, or gap; to hear the past and present withdrawal of sound from an ecosystem and its soundscape, through ecological destruction?

And once we have listened to this silence, how best to memorialise this loss? In a series of works focusing on what Dugal McKinnon has termed “ecological silencing” I explore the possibility for practice based research to investigate the lost birdsongs found within New Zealand colonial narratives, asking what it might mean to re-collect, through interventions into archival records, notation, and other material traces, the songs of bird species lost before the invention of recording technologies, and then to situate these lost songs back into hearing, including placing them back within what might be now listened to, and represented in phonographic or field recording practice, as the 'natural' soundscap

TRACES IN/OF/WITH SOUND: AN ARTIST'S EXPERIENCE OF AUDIO-VISUAL SPACE

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ABSTRACT

Traces in/of/with Sound[1] was an audio-visual performance series, instigated as part of the author's practice based PhD research into the process of sound arts practice (Garrelfs 2015). The initial idea for the project resides in the realm of visual music and an interest in the influence that the relationship between sound and image has on the music that is produced within performance. The research employed a modular methodology that includes creative practice as a key space - or in-vivo laboratory - in which the process of this practice can be studied.



Figure 1: Stills from a later version of "Traces in/of/with Sound"

As a piece of creative work, *Traces in/of/with Sound* made use of a field of juxtapositions: a projection of recorded and digitally mediated drawings with improvised and digitally processed voice; notions of archetypes across sound and vision; a range of complex conceptual concerns with a performative experience. In its inception, several strands of thought combined. These included recognising a similarity between Norman McLaren's images (Barbeau 2005) and some of the author's drawings (see Figure 2); a concern with movement - as explored in her previous locative mobile phone pieces *override* (2011) and *In A Day's Work* (2011) - that transferred onto the relationship between the eye's movement and still images (Brown 2006). In addition, the notion of archetypes expressed through line drawings (Ingold 2007) and vocal expression met with digital processing techniques.

Between 2011 and 2013, six performances and one installation took place, each with a different audio-visual spatial configuration, ranging from mono sound / single screen video to eight-channel sound / two screen video. Each of these versions brought with it adaptations of the core material, as a response to the preceding incarnation.

What remained stable was the method of performing: sound used live improvised voice, manipulated and diffused across a multichannel system (where applicable) via Cycling 74's Max software. This sound material was created as an improvised response to a pre-prepared "film" of digitally manipulated drawings using Adobe Creative Suite packages Photoshop and Premiere. As the series developed, working within this complex field of juxtapositions led to a change of focal points. Whilst the project began by essentially considering movement and a contrapuntal relationship between sound and vision as a property of time, it shifted to an exploration of a joint audio-visual spatiality, understood as a perceptual experience established by the interlaced movements of both sound and vision.

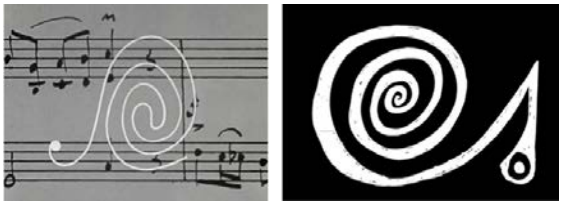


Figure 2: On the left is depicted one of Norman McLaren's images (Barbeau 2005). The image on the right is an early still image from "Traces in/of/with Sound"

This paper will chart the development of the piece over a two-year research period from 2011-2013, including the presentation of relevant stereo extracts and the work's relationship with theoretical concerns and wider creative praxis. Within this narrative it will pay particular attention to the author's emergent experience of a speaker & screen based audio-visual spatiality. Some conclusions as to strategies that may promote experiential coherence or disunity in the perception of an audience with respect to screen based audio-visual space will also be put forward.

ACKNOWLEDGEMENTS:

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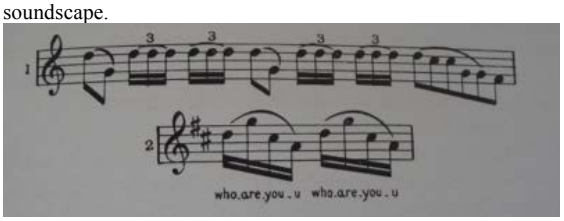
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Huia Transcriptions: re-collecting Colonial era witness accounts of extinct birdsong

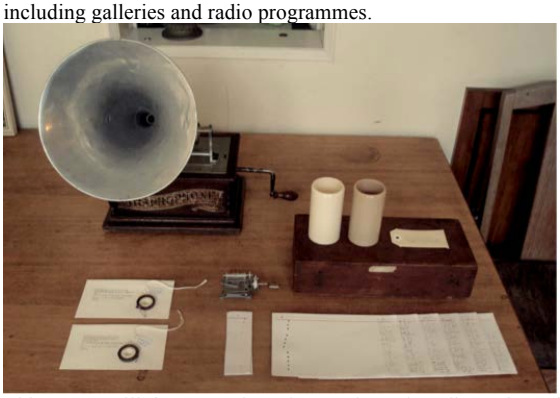
Sally Ann McIntyre
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ABSTRACT

In classical acoustic-ecological conceptions of the soundscape, the technological preservation of a sound mark might be understood to positively relate to the preserving of memory of place. But what of the sounds beyond (recorded) memory, that are already missing? Might there be value in suspending the fantasy of a natural plenitude of sonic fecundity, and its status as a potential recorded totality, to adequately acknowledge this haunting, or gap; to hear the past and present withdrawal of sound from an ecosystem and its soundscape, through ecological destruction? And once we have listened to this silence, how best to memorialise this loss? In a series of works focusing on what Dugal McKinnon has termed "ecological silencing" I explore the possibility for practice based research to investigate the lost birdsongs found within New Zealand colonial narratives, asking what it might mean to re- collect, through interventions into archival records, notation, and other material traces, the songs of bird species lost before the invention of recording technologies, and then to situate these lost songs back into hearing, including placing them back within what might be now listened to, and represented in phonographic or field recording practice, as the 'natural' soundscape.



These works have re-collected the distress calls of birds such as the extinct huia, by placing notation of their songs written in the late 1800s back into the audible world, situating them within mnemonic technologies that are not current, but classifiable as 'heirloom' media. These technologies are not focused on as technological fetishes or museum pieces, but as functional material witnesses to the birds themselves, and include the wax cylinder and the programmable music box, they have facilitated the making-audible of the bird songs and have allowed them to be placed back within the soundscape of New Zealand bird sanctuaries, and other environments including galleries and radio programmes.



This paper will focus on these issues through a discussion of the works, and also explore how they articulate a tension between the need to memorialise such loss in the veneration of individual species, and the recognition that the lost remain as traces within the wider physical and acoustic environment, drawing together the current soundscape and museums and archives, notation, written narratives and other recordings, in an attempt to historicise the practice of listening to nature. Just as George Gibbs relates in his book *Ghosts of Gondwana* that the Moa is still visibly present in the New Zealand landscape through the defence mechanisms of the Lancewood tree's juvenile stage, might we also be able to locate a sonic equivalent in the after-echoes of the songs of the Huia within the songs of extant species? Perhaps, as John C. Ryan suggests in his essay, *Why Do Extinctions Matter?* "A more tenable ecological conceptualisation of mourning needs to consider connectivity, rather than unified subjectivity, as a tool for exploring the deep channels of grief over the loss of the more-than- human."

ON SOLO – A PROGRESS REPORT

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ABSTRACT

This paper reports on an ongoing research project around Karlheinz Stockhausen's historical work Solo (*Solo, für Melodie-Instrument mit Rückkopplung* 1965-6).[1] Together with Dr. Juan Parra Cancino from ORCIM Ghent, we are teasing out the consequences of the (nowadays common) software replacement of the elaborate tape delay system that was used at the time of the work's inception. Many of the technical elements (e.g. in and output levels, delay feedback level, output levels) were operated as prescribed by the score by no less than three technicians. These roles have now been integrated into software patches (e.g. MaxMsp or PD (Sluchin 2000)) but also an app for i-Phone and pad (Petrolati 2016). Software approaches integrate and automate the score to different levels, with one version going as far as integrating the soloist's sound source into the digital domain (Esler 2006).

Even thought this work's score is strongly prescriptive, like many of the composer works, performances and recordings of Solo have been augmented in different ways, most notably the addition of a layer of electronic music in the work's first recording with trombonist Vinko Globokar (from 1969). Another freedom provided in the score for this work is the choice of different timbral modes (e.g. a choice of mutes in the case of brass players).

Where the proto-affordance of analogue tape is reproducibility, the proto-affordance of a computer simulating a tape system is much richer, with its ability to compute and facilitate interaction. At the same time, a computer these days implies a network, commonly the Internet. To simply reduce the computer to a simulator of older technologies and not exploring what digital technology affords seems to go against the grain of this influential work, and of electronic music performance.

In our research project we attempt to pull apart the distinct layers of input (soloist), delay system (software), level of control and loudspeaker system output (usually four channels). We have produced some concerts with these four layers performed either in the same venue, or in different venues (and even countries).

Computer network connections are used to share timing information and distribute audio between different performers and audiences. For instance, by having the soloist in a different venue than the public, the mixer (i.e. sound projection and level control) and loudspeakers, we can query the technological and human agency, of each of these steps.An additional research element of using network audio connections is that, not dissimilar to the build-up of noise in a tape-loop system and other analogue tape trademarks, the digital sonic artifacts (e.g. network 'sounds') become an audible element of the performance. In a way, those instances give a voice to the, ordinarily auditory hygienic, digital processing.

In addition to a series of concerts, the project comprises of interviews with some of the key players in early performances and integration with our respective research interests: the role of loudspeakers (Mulder 2013) and the performance practice of computer music (Parra Cancino 2014). The academic outputs of this project consist of a number of conference performances/papers and a journal article.

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LISTENING FOR PRESENCE

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ABSTRACT

The question of sound as experience is critical to discussions about environmental listening. I have come to think of sound as a viscous material, a vibrating energy field that has texture and density and a physicality that is unlike most other media.
I arrived at this view through a combination of several experiences and practices. The first being a process of duration environmental listening and the second, an invitation to be part of an Australian indigenous dreaming ceremony at Bundanon in NSW. These experiences brought me to a point of knowing that everything is part of an N dimensional vector field - where by energy fields can be attracted together to form a presence in the world. I came to think of sound in these terms. Sources move, perhaps in relation to each other and in relation also to environmental forces, all making up a manifold and complex morphology, a rich and largely invisible to me interconnectedness.
This interconnectedness is an experience I have sought increasingly deeply through durational listening where I ask myself, "what does it mean to be truly present", is it possible to continuously deepen that sense of presence through repeated practice?

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Sonic Placemaking in the American Southwest: The Listen^{n} Project

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School of Music

ABSTRACT

Sound gives life to our environment. Sound heightens our experience of place. Initiated in 2013 by composer Garth Paine, the Listen^{n} project capitalizes on the vibrancy of environmental sound in the American Southwest. As indicated by its title and superscript n, the project explores multiple ways of listening, promoting listening in physical and virtual locations. It is collaborative and interdisciplinary, combining science, media art, technological innovation, and community engagement.
This presentation provides insight into Listen^{n}’s fieldwork in parks of the American Southwest to create, with ambisonic recording technology, the largest online database of geo-located field recordings representing these places.

Light is shed on a series of compositions crafted from these recordings and community workshop. Attention is also drawn to virtual reality experiences of place created through the pairing of desert sounds with 360-degree photographic panoramas of the sounds’ place for display on the Oculus Rift headset (EcoRift) to allow distant communities to be remotely present in a landscape.
Listen^{n} builds on acoustic ecology-based research and art and sound mapping (*Biosphere Soundscapes*, *Nature Sound Map*, *Living Symphonies*), but it is unique in its scope, extended time span, multi-platform design and engagement of local and global communities.

WHAT CAN WE LEARN FROM LISTENING TO NATURE?

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ABSTRACT

Natural soundscapes provide a wealth of information, both to the casual listener and the research scientist. This ranges from aesthetic pleasure, to identification of species repertoire and behaviour, and measures of ecosystem health.

When we consider repertoires in an evolutionary context, we can understand sonic strategies as not only shaping behaviours and survival adaption, but being fundamental to speciation itself. This has been documented in certain insects, and can be speculated upon in higher animals.

This leads to an enquiry into the possible role of sonic communication in hominin development, and the suggestion that rhythmically synchronised communication (music) may be viewed as a biological rather than cultural phenomenon, unique to humans, of great antiquity, the result of which has been a gradual development of the higher brain functioning and eusociality distinctive to the hominin lineage.

While sound and acoustic communication may have shaped us, we are changing the natural soundworld at the very moment we are beginning to study and appreciate its richness. Natural sonic environments are coming to be seen as in need of preservation, and healthy soundscapes important for our wellbeing too.

Listening has been described as the universal sense. If music is culturally important as more than simply entertainment, personal taste or product, then it needs to address the challenges of our time; to help us find our human place in the natural world.

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2.

Sonic Environment in Vanuatu: Exploring Water Music TRACES

Sandy Sur
Leweton Cultural Group

ABSTRACT

Sandy Sur's research focuses around the Water Music of Vanuatu and its connection to the environment. Water connects everything on earth and is essential for survival. At a time when the world is facing so many environmental challenges it is more important than ever before to deeply understand the role of water in our life. Understanding the sound and rhythm of Vanuatu Water Music allows us to explore the environment in new ways and develop a deeper understanding of the role sound plays in the environment. The Water Music of Vanuatu is site-specific and deeply inspired by the surrounding environment. This inspiring tradition is now evolving in response to rapidly changing climates that are affecting island communities.

Sandy is one of the only people in the world holding the knowledge to lead research on water music and over the last decade he has directed a wide spectrum of research projects designed to bring water music to the world. Sandy's research showcases this tradition as a way for understanding the environment at a time when we urgently need to listen to nature. His research is realised as live performances, films, recordings and web based media designed as tools for reaching the world.

Sandy Sur is the Manager of the Leweton Cultural group. While the water music of Vanuatu is a once-in-a-lifetime performance that needs to be seen (and heard) to be believed – the Leweton Cultural group deliver a range of customary artisanal performances and workshops including dances, weaving, carving, mixed-media/found objects, environmental art, and instrument-making. Sandy coordinates residencies for the group at international festivals and events with deep connections to people of place. The Leweton Cultural Group has stunned audiences at World Expo 2008 in Zaragova, several European Union diplomatic functions, the Rainforest World Music Festival in Borneo, the Bellingen Global Carnival, Queensland Music Festival, and the Floating Land Festival in Australia and the Lukaotem Gud Santo Festival in Vanuatu. Sandy is a ranked man and leader from Merelava in Vanuatu. A skilled carver, weaver, craftsman, and curator Sandy is adept at using elements of the natural world as well as found objects and ubiquitous technical items such as wire, rope, etc to create functional and beautiful objects.

This research paper explores the sonic environments of Vanuatu and the role of Water Music.

The Piano Mill

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ABSTRACT

It has been suggested that as many as three out of four Australians may have had a piano by the end of the nineteenth century. (Rose 2008). In fact, between 1788 to 1888, Australia possessed more pianos per head of population than any other country. But how did the parlours, pianos, and their players, translate to the Queensland bush? The Piano Mill is in part an investigation of this story that connects place and music.

The Piano Mill is a purpose built structure and musical instrument in the Granite Belt near Stanthorpe, Qld. A collaboration between architect Bruce Wolfe and composer Erik Griswold, this structure houses 16 found pianos over 2 levels. The audience listens to the "mill" from outside, unable to view the workers (16 pianists) as they interpret Griswold's score, *Alls Grist That Comes to the Mill*

The Piano Mill, examines the pioneering history of Australia through the gaze of now discarded pianos; celebrating place, community, environment and above all, listening. This was constructed as a one-off performance event, probing ideas of nostalgia, transformation of land and function and the sheer joy of creativity.

This presentation will weave together the perspective of the architect, the composer, audience members, performers, and directors, to gain a kaleidoscopic view of the event that was the Piano Mill. As a piece of architecture, a sounded building, and an integral part of a larger artistic vision, this multi-dimensional approach is necessary to gleen something of the intense joy that has gathered in and around the mill.

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Exploring Internet Environments in Sound Arts

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ABSTRACT

Contemporary environmental sound art is often linked with geographically dispersed local cultural practice and/or natural environmental sound. And recently explored in tandem, the development of telematic sound arts are dominated by linking electroacoustic music studios and/or concert halls. Yet urbanised life often involves inhabiting bodies, local environments *and* digitally interconnected global environments including people, computer-based agents, and aggregates real-time informational data streams.

Current eResearch suggests this new environment will increase through: connectivity, greater bandwidth and processing power; smart/embedded technology and the Internet of Things; artificial intelligence and automated decision-making; data streams and making knowledge out of information with machine learning. What role can sonic art practices play in navigating increasingly complex relationships, represented particularly in multiple aggregated information flows?

Recent work on radical embodied cognition, reacting against older computational views of intelligence using symbolic processing and absent bodies, suggests human cognition is situated and time-pressured, is environmentally relational, used for action, and that much offline cognition is body-based. Accordingly, while current data streams, such as news feeds, can be rendered visually, we partly interpret these through mood, metaphor and movement, similar to music reception.

A meeting point for telematic sound arts and networked life that practitioner might further explore is in affective composition/performance models coupled with the sonification of real-time information streams. This involves further amalgamating research on the affective dimension of electroacoustic music with real-time data sonification techniques to extend performance-based electroacoustic music languages. And this process could further be automated with the integration of emerging research that adopts machine learning techniques in the gestural mapping of sound, together with the application of intelligent-agent decision making technology used in sound arts, thus enhancing rendering efficiency.

This allows for the sonic exploration of our place in a matrix of increasingly networked and non-liner relationships with dynamic meanings through creating new knowledge. And we may find new patterns in data streams only possible through aural means, allowing us reimagine our networked presence and relationship with place.